R Practice Module 1

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Introduction:

The Census Income dataset, commonly referred to as "adult.data" comprises 32,560 entries, each providing detailed information about an individual. This dataset offers a comprehensive snapshot of various socio-economic attributes and demographic factors for a diverse population. The dataset covers a range of features, providing insights into the demographic, educational, occupational, and economic characteristics of individuals.

```
adult.data <- read.csv("adult.data")
```

The data provided has been assigned different names to improve clarity and understanding, resulting in changed column names.

```
library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union

# Renaming Column X39
adult.data <- adult.data %>%
  rename(Age = X39)

# Renaming Column X77516
adult.data <- adult.data %>%
  rename(Fnlwgt = X77516)

# Renaming Column State.gov
adult.data <- adult.data %>%
  rename(Workclass = State.gov)

#Renaming Column Bachelors
adult.data <- adult.data %>%
  rename(Education = Bachelors)
```

```
# Renaming Column X13
adult.data <- adult.data %>%
  rename(Education_num = X13)

# Renaming Column Never.married
adult.data <- adult.data %>%
  rename(Marital_status = Never.married)

# Renaming Column Adm.clerical
adult.data <- adult.data %>%
  rename(Occupation = Adm.clerical)

# Renaming Column Not.in.family
adult.data <- adult.data %>%
  rename(Relationship = Not.in.family)

# Renaming Column White
adult.data <- adult.data %>%
  rename(Race = White)

# Renaming Column Male
adult.data <- adult.data %>%
  rename(Sex = Male)

# Renaming Column X2174
adult.data <- adult.data %>%
  rename(Capital_gain = X2174)

# Renaming Column X0
adult.data <- adult.data %>%
  rename(Capital_loss = X0)

# Renaming Column X40
adult.data <- adult.data %>%
  rename(Hours_per_week = X40)

# Renaming Column United.States
adult.data <- adult.data %>%
  rename(Native_country = United.States)

# Renaming Column X..50K
adult.data <- adult.data %>%
  rename(Income = X..50K)
```

Renaming Text:

In the data frame, certain columns were assigned ambiguous names. Consequently, the gsub() function was employed to modify these names, enhancing their clarity and facilitating improved comprehension within the dataset.

```
# In Marital_status Column Married-AF-spouse changed to Married-Armedforce-
spouse.
adult.data$Marital_status <- gsub("Married-AF-spouse", "Married-Armedforce-
spouse", adult.data$Marital_status)

# In Marital_status Column Married-civ-spouse Changed to Married-Civilian-
spouse
adult.data$Marital_status <- gsub("Married-civ-spouse", "Married-Civilian-
spouse", adult.data$Marital_status)

# In Work class Column Self-emp-not-inc changed to Self-employed
adult.data$Workclass <- gsub("Self-emp-not-inc", "Self-employed",
adult.data$Workclass)

#. In Occupation Column Adm-clerical changed to Office Clerk
adult.data$Occupation <- gsub("Adm-clerical", "Office Clerk",
adult.data$Occupation)

#In Occupation Column Exec-managerial changed to Manager
adult.data$Occupation <- gsub("Exec-managerial", "Manager",
adult.data$Occupation)

# In Occupation Column Handlers-cleaners changed to Cleaner
adult.data$Occupation <- gsub("Handlers-cleaners", "Cleaner",
adult.data$Occupation)

#In Occupation Column Tech-support changed to Tech-expert
adult.data$Occupation <- gsub("Tech-support", "Tech-expert",
adult.data$Occupation)
```

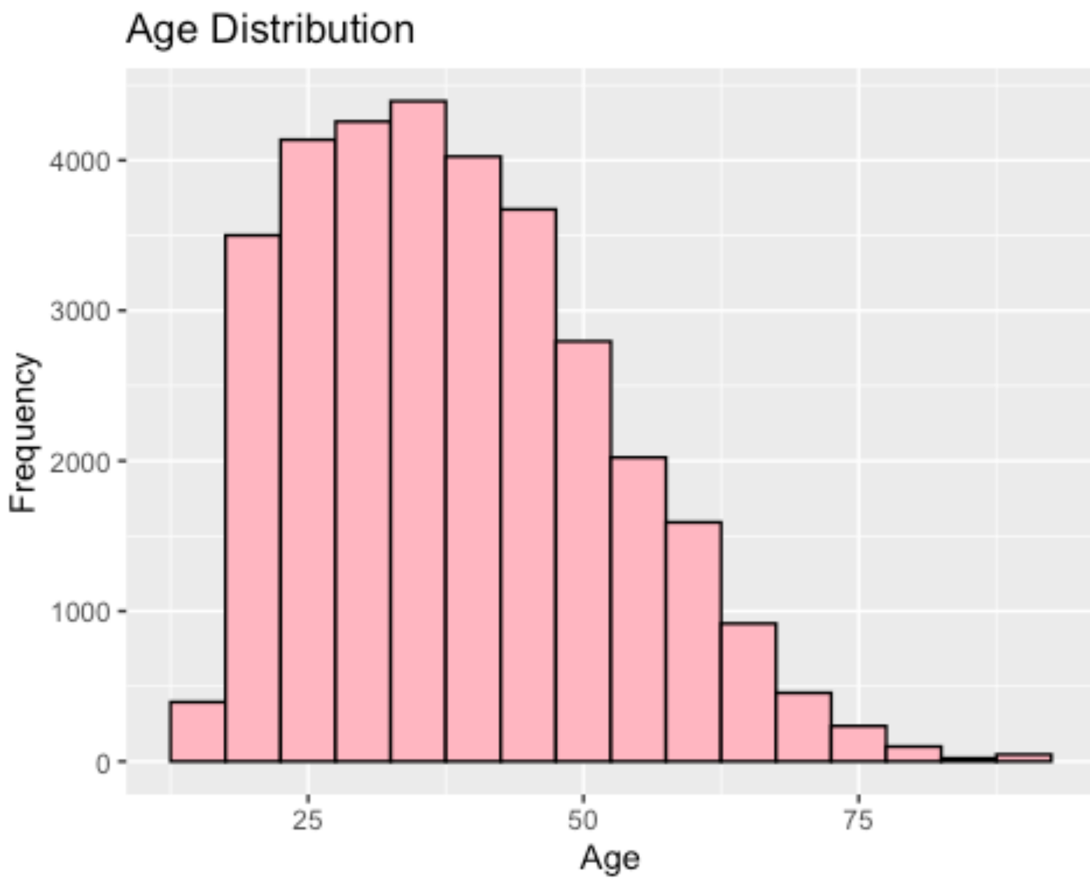
Dropping Columns:

In order to make the dataset easier to understand and work with, certain columns have been chosen for removal. These columns were found to be unclear or added unnecessary complexity to the dataset, so they were dropped to simplify the information and improve overall clarity.

```
# Dropping Education_num
adult.data <- subset(adult.data, select = -Education_num)

# Dropping Fnlwgt
adult.data <- subset(adult.data, select = -Fnlwgt)

#Dropping Capital gain
adult.data <- subset(adult.data, select = -Capital_gain)

#Dropping Capital Loss
adult.data <- subset(adult.data, select = -Capital_loss)
```

Age Distribution Graph

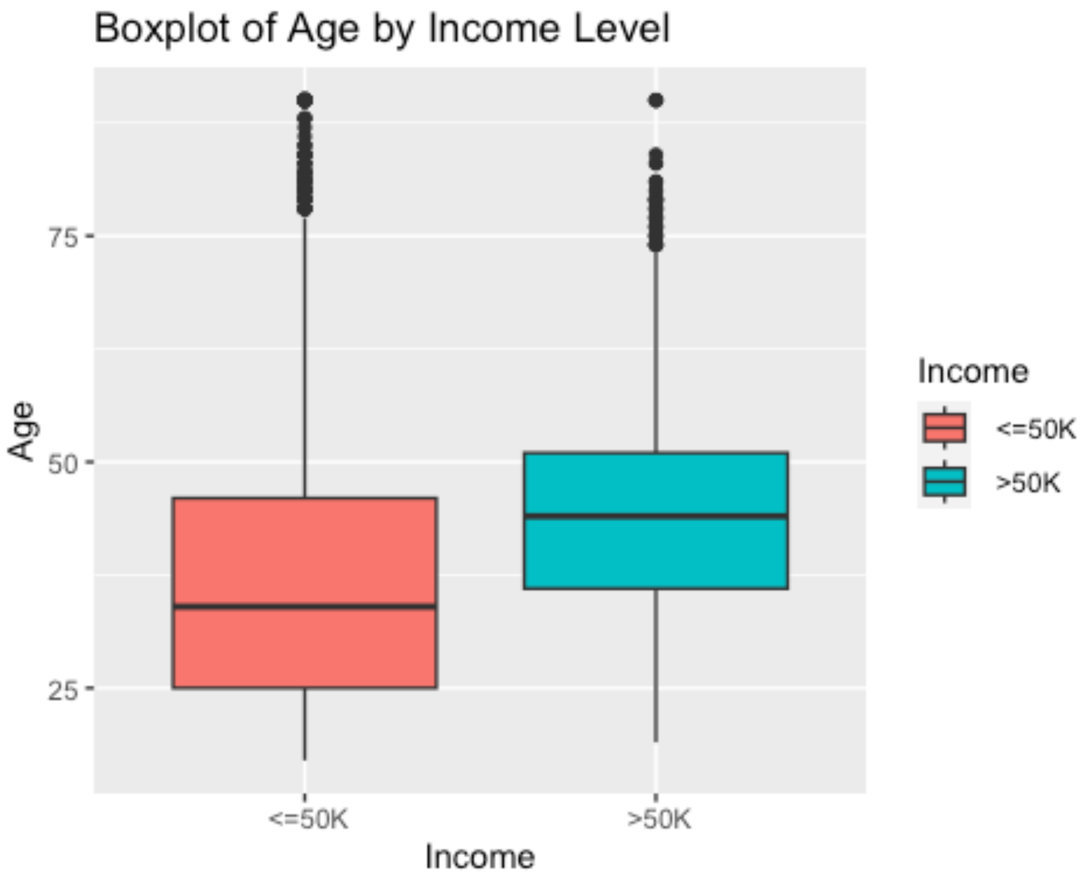
```
library(ggplot2)
ggplot(adult.data, aes(x = Age)) +
  geom_histogram(binwidth = 5, fill = "lightpink", color = "black") +
  labs(title = "Age Distribution", x = "Age", y = "Frequency")
```



The Age Distribution Graph serves as a visual representation of the age distribution within the dataset, offering insight into the demographics of individuals included in the data. Notably, the graph highlights a concentration of individuals aged between 25 to 50, indicating a higher population within this age range. This understanding of age distribution is crucial for analyzing age-related patterns or trends within the dataset. Visualizations like this provide a clear overview, facilitating the identification of significant patterns or outliers present in the data.

Age Distribution By Income

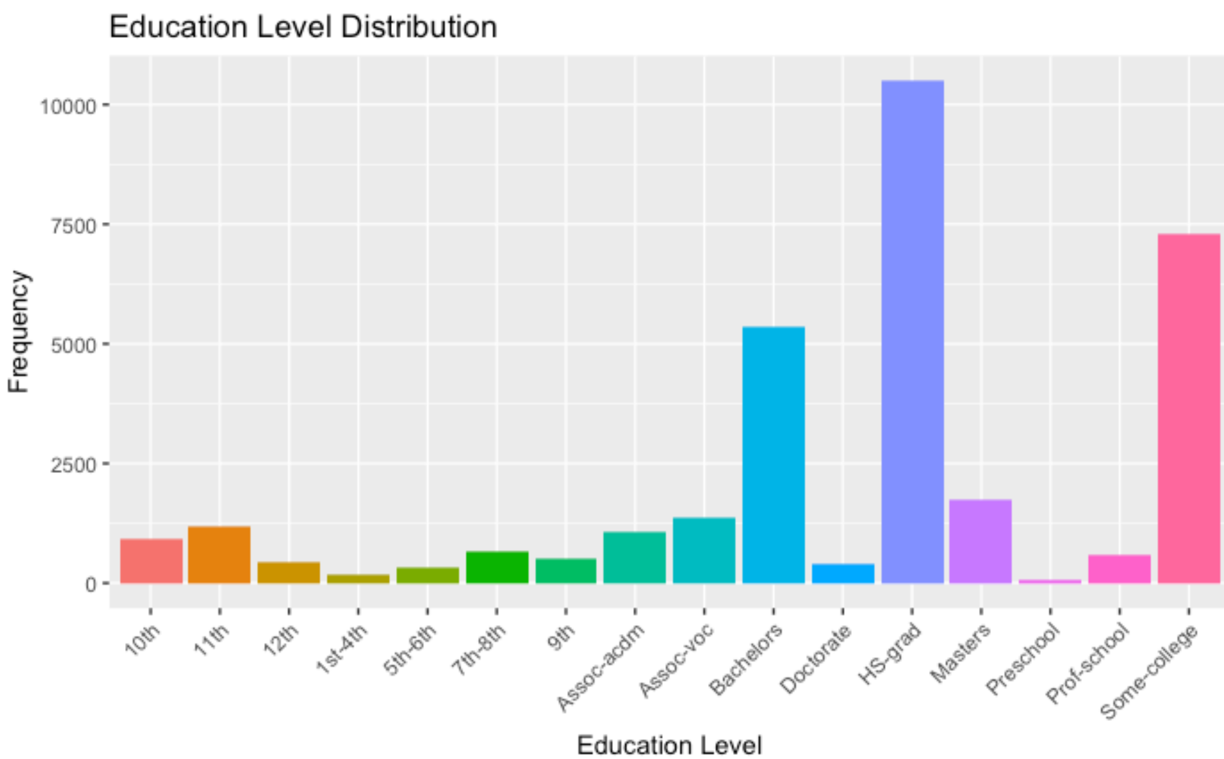
```
ggplot(adult.data, aes(x = Income, y = Age, fill = Income)) +
  geom_boxplot() +
  labs(x = "Income", y = "Age", title = "Boxplot of Age by Income Level")
```



The graph illustrating age distribution based on income levels has been showcased to visually depict the relationship between age and income within the dataset.

Education Level Distribution

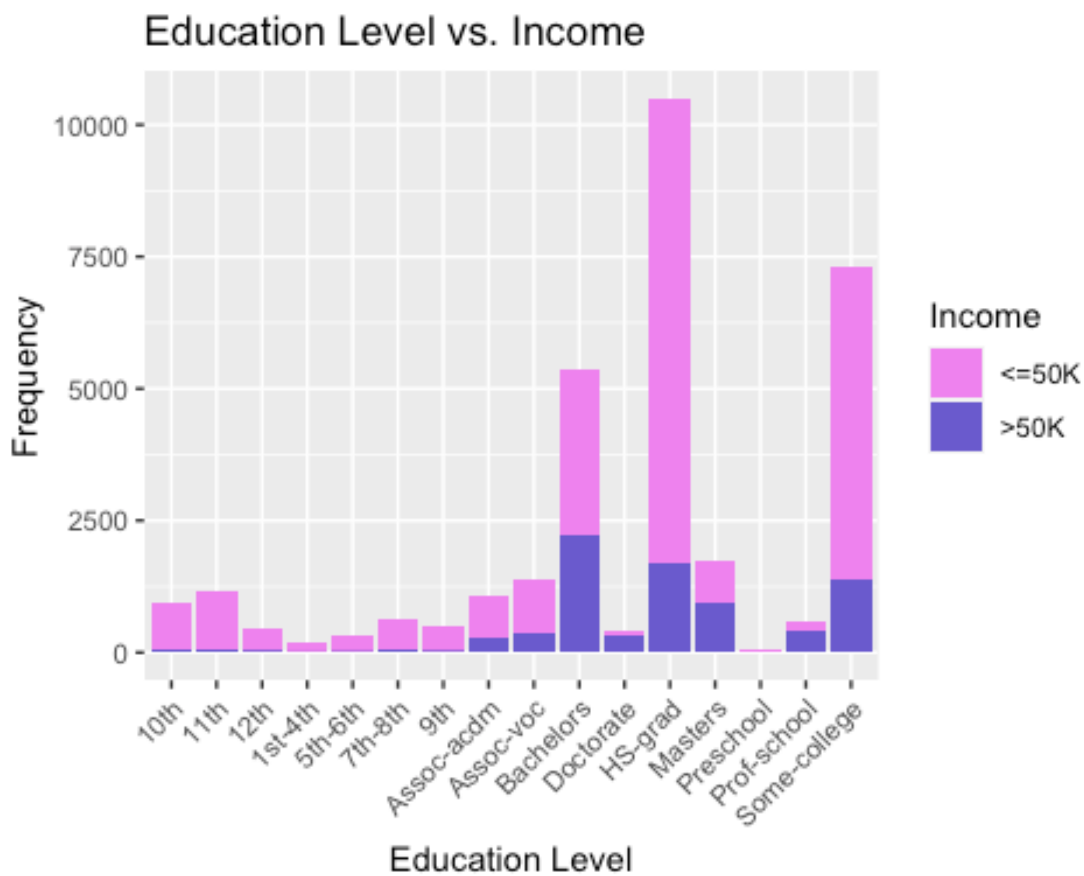
```
ggplot(adult.data, aes(x = Education, fill = Education)) +
  geom_bar() +
  labs(x = "Education Level", y = "Frequency", title = "Education Level
Distribution") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1), legend.position =
"none")
```



The education distribution plot demonstrates that the population with a high school graduation (HS-grad) education level is the most prevalent, followed by individuals with some college education. This visual representation highlights the dominant presence of individuals with a high school diploma in the dataset, with the next largest group being those who have completed some college coursework.

Education By Income

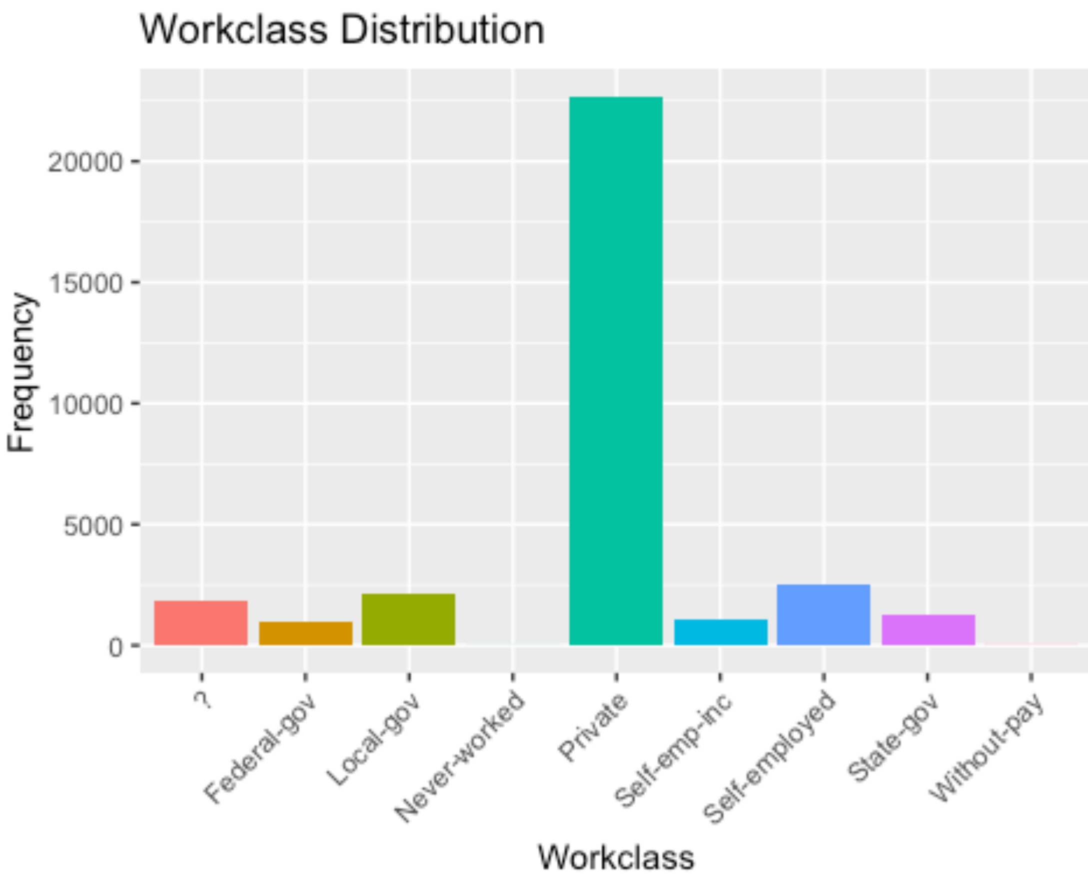
```
ggplot(adult.data, aes(x = Education, fill = Income)) +
  geom_bar() +
  labs(x = "Education Level", y = "Frequency", title = "Education Level vs.
Income") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  scale_fill_manual(values = c("violet", "slateblue"), labels = c("<=50K",
">50K"))
```



The plot depicting education level by income reveals that among individuals with a high school graduation (HS-grad) education, less than 1300 individuals earn a salary of $50,000 or more.

Workclass Distribution

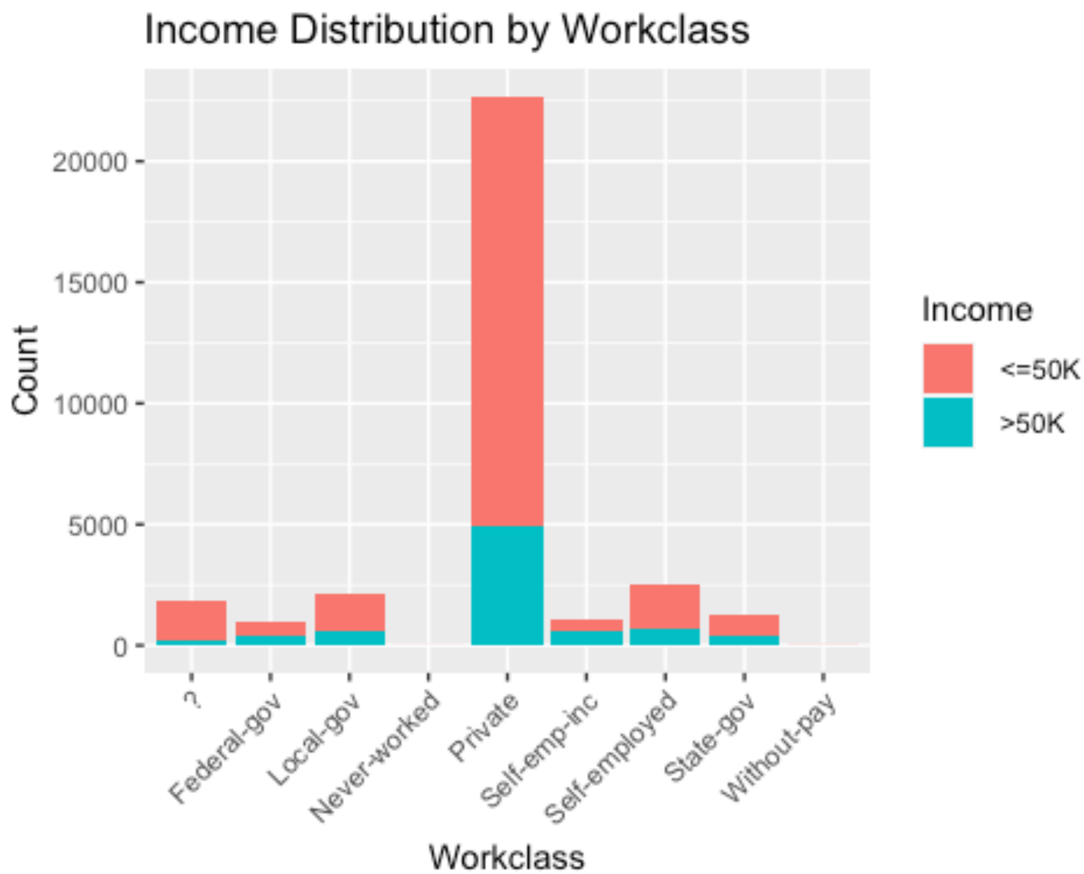
```
ggplot(adult.data, aes(x = Workclass, fill = Workclass)) +
  geom_bar() +
  labs(x = "Workclass", y = "Frequency", title = "Workclass Distribution") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1), legend.position =
"none")
```



In the work class distribution plot, it is evident that approximately 22,000 individuals are employed in the private sector. This indicates that a significant portion of the dataset is comprised of individuals working within the private industry.

Work class Distribution By Income

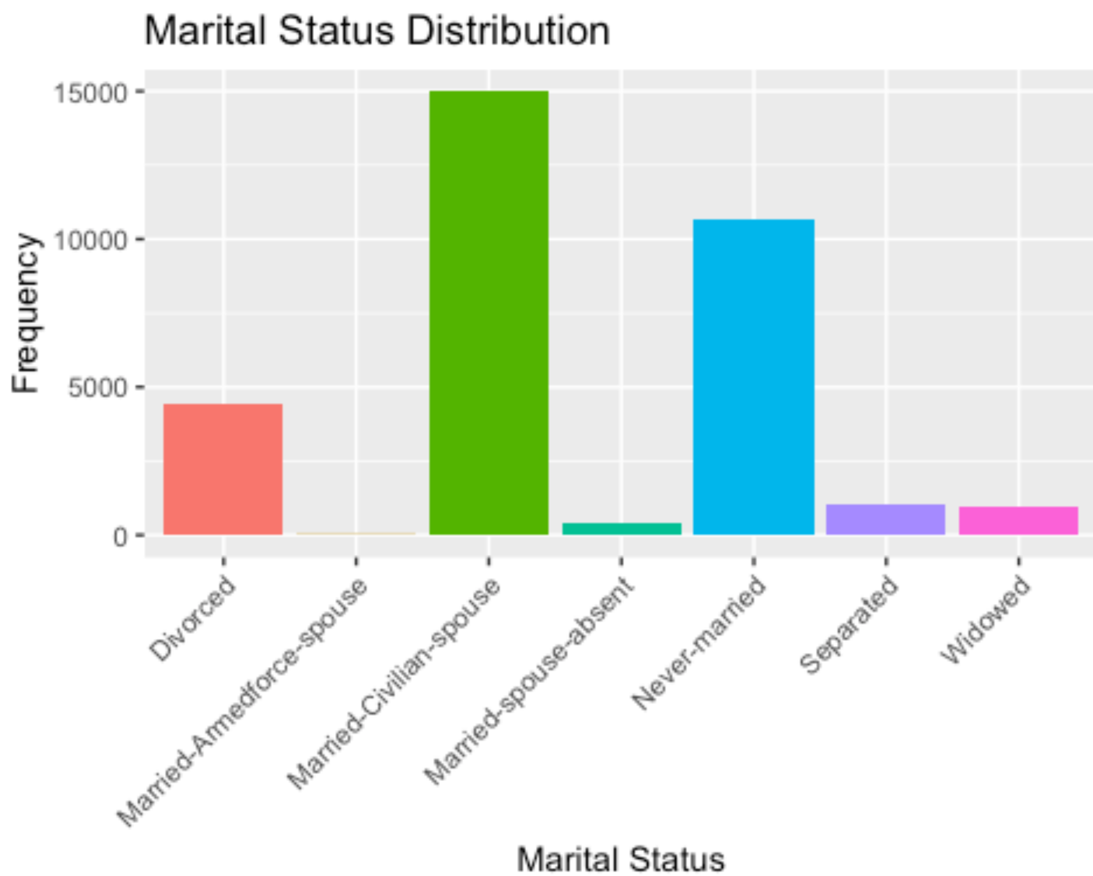
```
ggplot(adult.data, aes(x = Workclass, fill = Income)) +
  geom_bar() +
  labs(x = "Workclass", y = "Count", title = "Income Distribution by
Workclass") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Around 5,000 individuals within the dataset are earning a salary that is equal to or higher than $50,000.

Marital Status Distribution

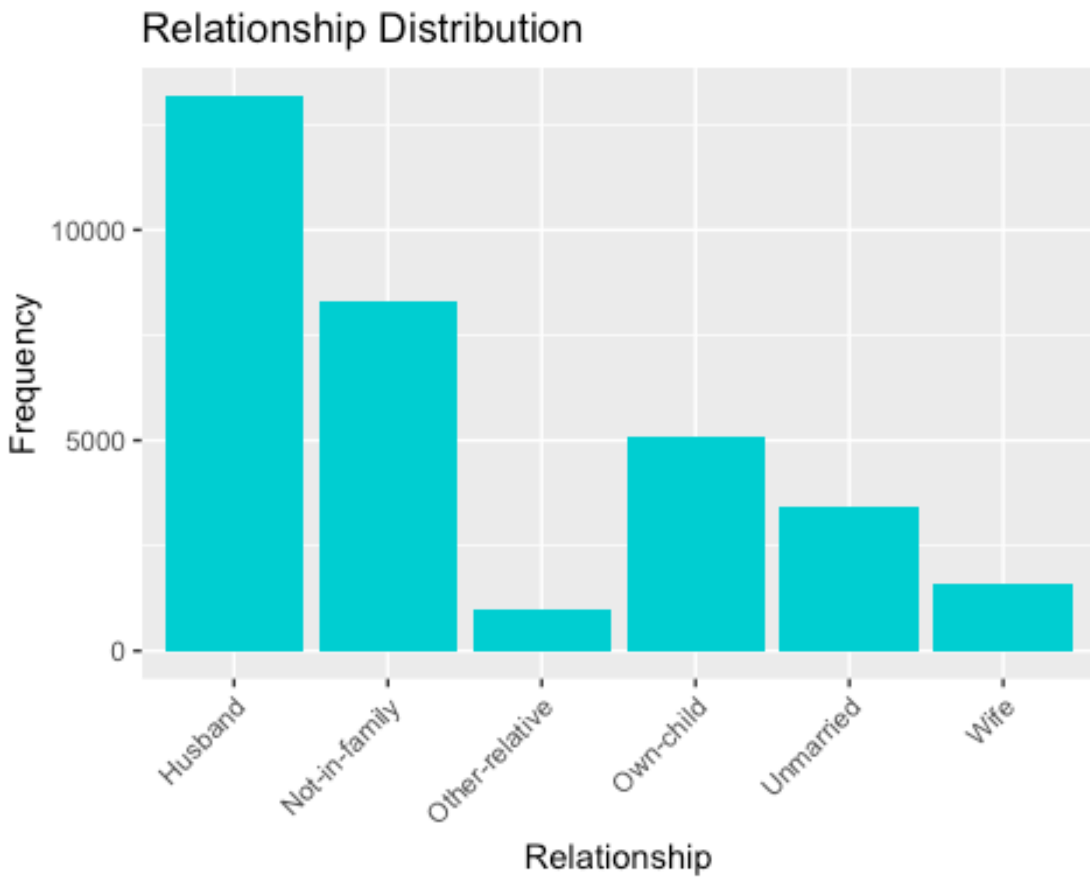
```
ggplot(adult.data, aes(x = Marital_status, fill = Marital_status)) +
  geom_bar() +
  labs(x = "Marital Status", y = "Frequency", title = "Marital Status
Distribution") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1), legend.position =
"none")
```



Within the marital status category, the data reveals that the largest segment consists of approximately 15,000 individuals who are married to a civilian spouse.

Relationship Distribution

```
ggplot(adult.data, aes(x = Relationship)) +
  geom_bar(fill = "darkturquoise") +
  labs(title = "Relationship Distribution", x = "Relationship", y =
"Frequency") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



The dataset includes more than 12,500 individuals who are classified as husbands, indicating a significant representation of married men within the data.

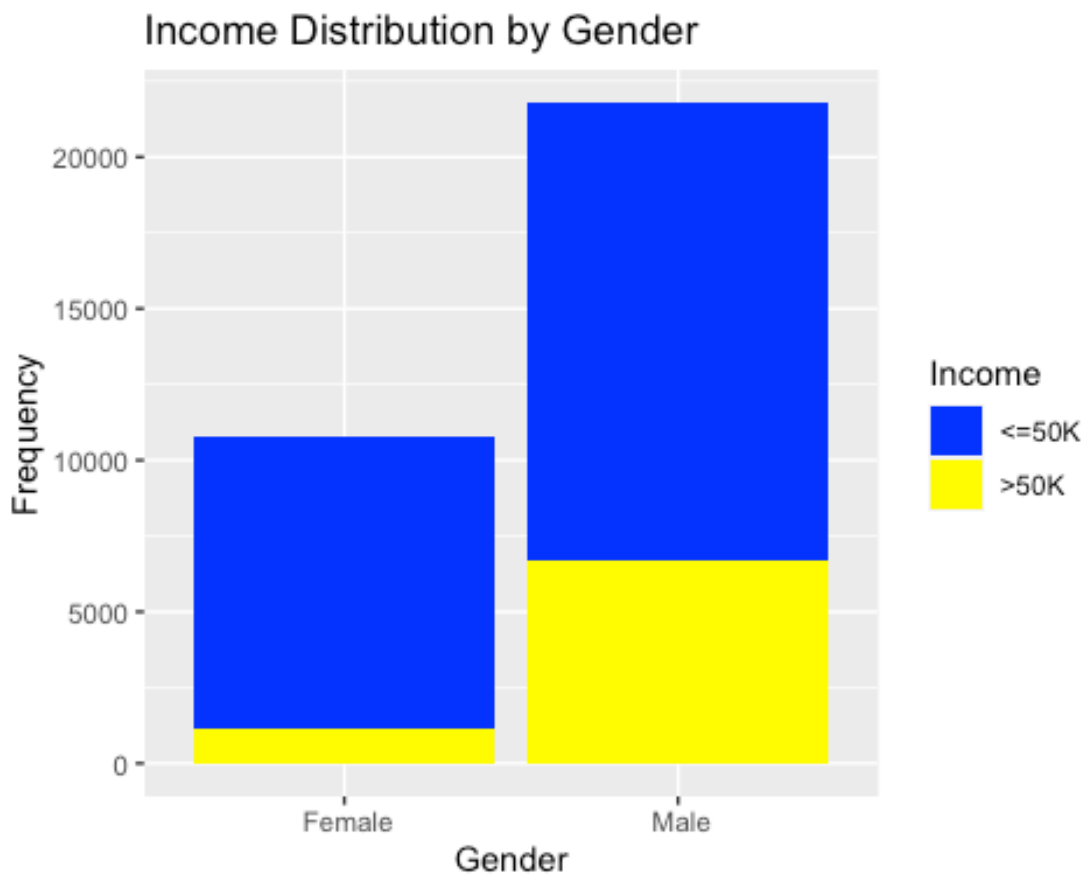
Gender Distribution

```
ggplot(adult.data, aes(x = Sex, fill = Sex)) +
  geom_bar() +
  labs(x = "Gender", y = "Frequency", title = "Gender Distribution") +
  theme(legend.position = "none")
```

In the gender distribution graph, it is observed that the population of males exceeds 20,000 individuals.

Gender Distribution by Income

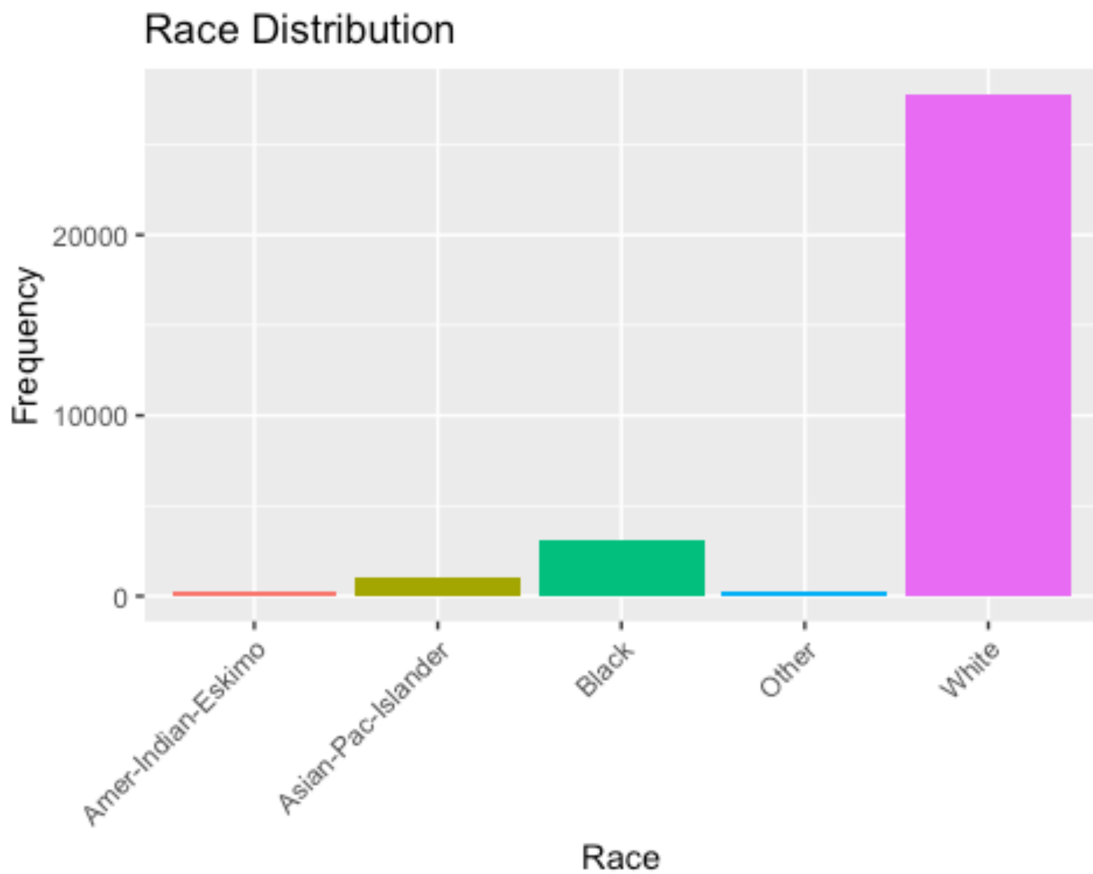
```
ggplot(adult.data, aes(x = Sex, fill = Income)) +
  geom_bar() +
  labs(x = "Gender", y = "Frequency", title = "Income Distribution by
Gender") +
  scale_fill_manual(values = c("blue", "yellow"), labels = c("<=50K",
">50K"))
```



This graph illustrates that among the population of over 20,000 males, approximately 5,000 to 7,000 individuals earn a salary exceeding $50,000.

Race Distribution

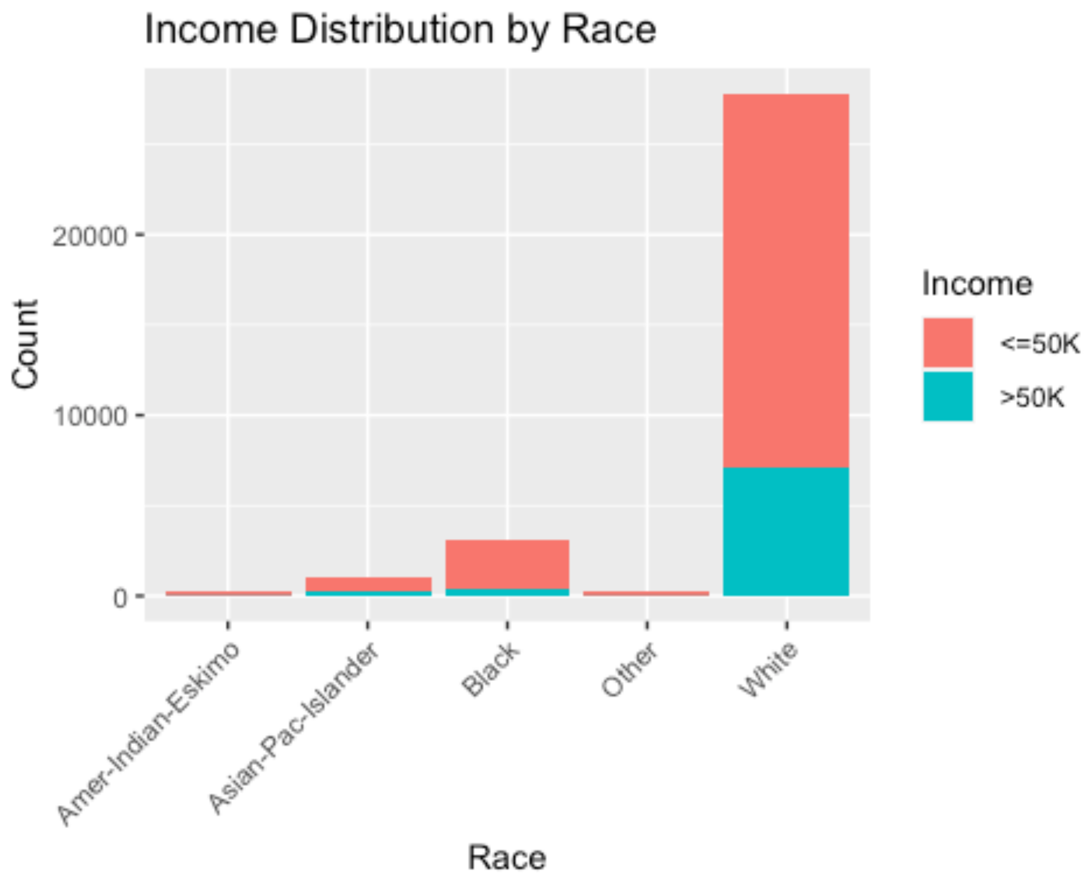
```
ggplot(adult.data, aes(x = Race, fill = Race)) +
  geom_bar() +
  labs(x = "Race", y = "Frequency", title = "Race Distribution") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1), legend.position =
"none")
```



In the race distribution graph, it is evident that the population of individuals belonging to a specific race exceeds 25,000, surpassing the count of individuals from any other racial group.

Race Distribution by Income

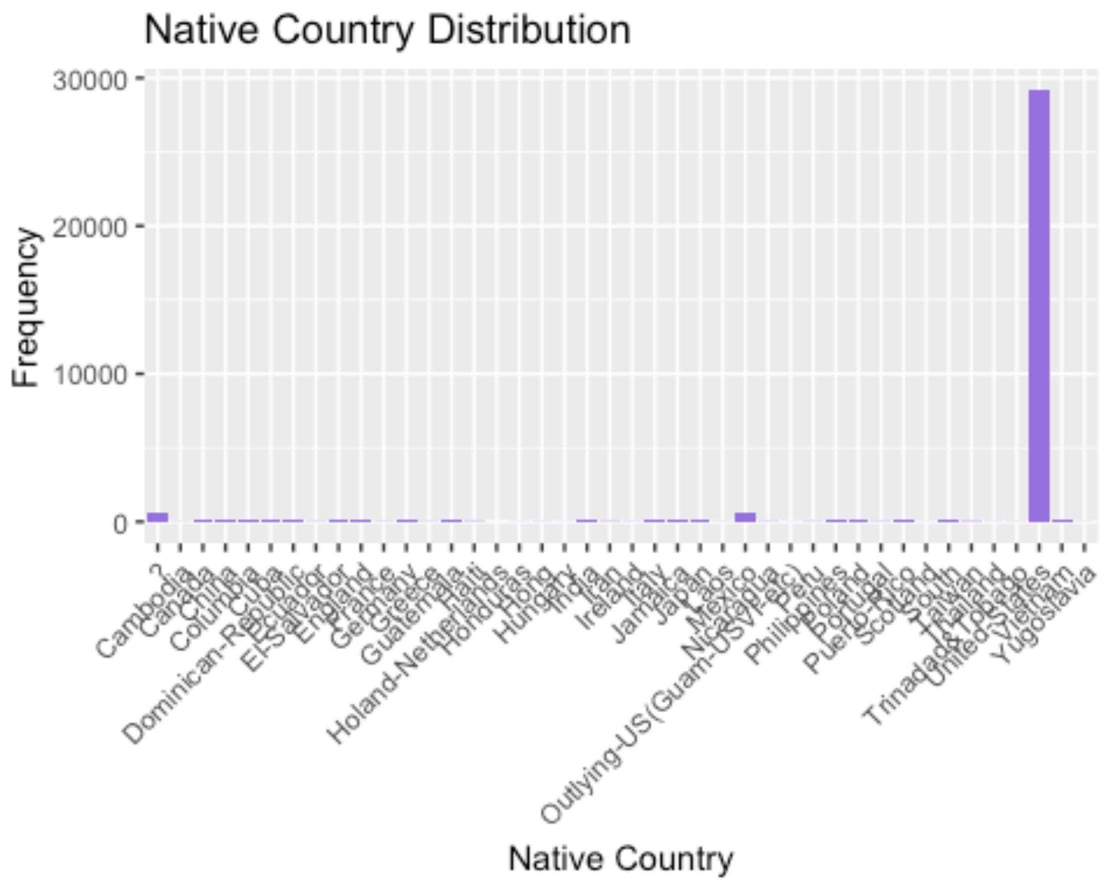
```
ggplot(adult.data, aes(x = Race, fill = Income)) +
  geom_bar() +
  labs(x = "Race", y = "Count", title = "Income Distribution by Race") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Among the white population, there are approximately 5,000 to 7,000 individuals earning a salary exceeding $50,000.

Native Country Distribution

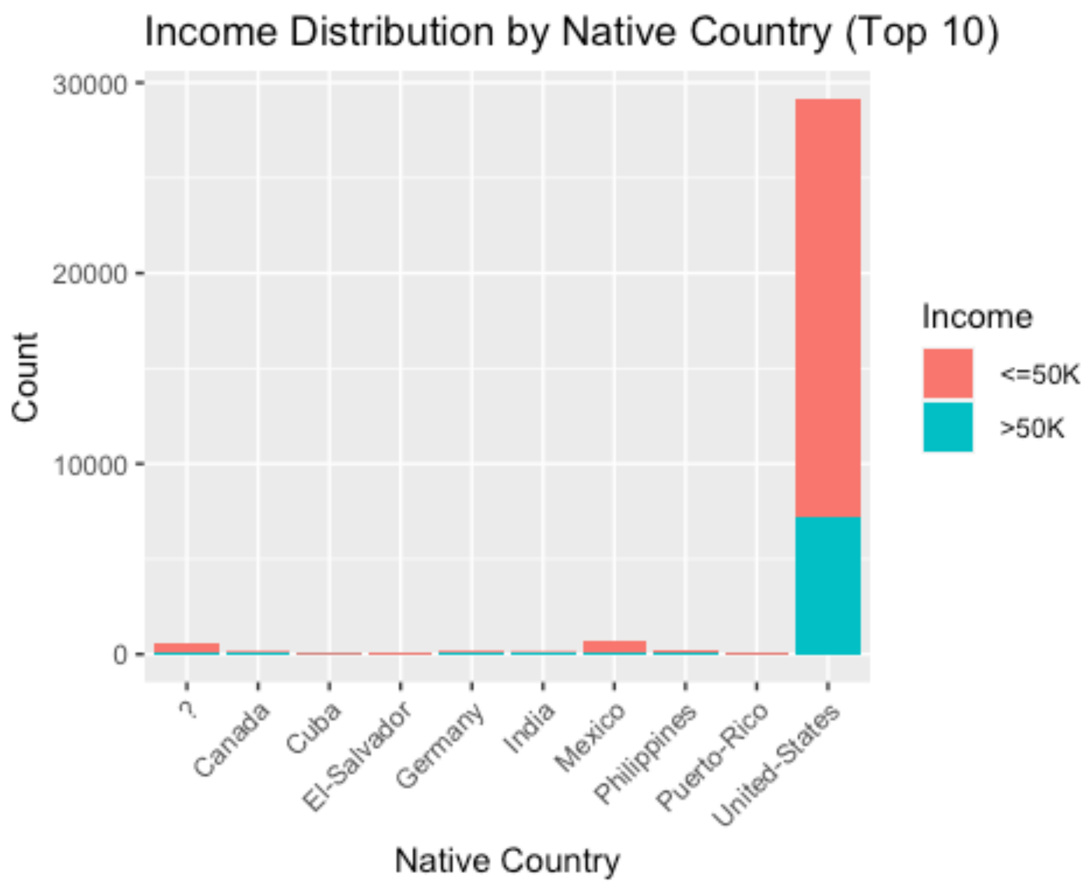
```
ggplot(adult.data, aes(x = Native_country)) +
  geom_bar(fill = "mediumpurple") +
  labs(title = "Native Country Distribution", x = "Native Country", y =
"Frequency") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Above Graph shows that the dataset was primarily collected within the United States.

Top 10 Native countries with income distribution

```
top_countries <- adult.data %>% group_by(Native_country) %>% summarise(count
= n()) %>% top_n(10, count)
ggplot(adult.data %>% filter(Native_country %in%
top_countries$Native_country), aes(x = Native_country, fill = Income)) +
  geom_bar() +
  labs(x = "Native Country", y = "Count", title = "Income Distribution by
Native Country (Top 10)") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



In the dataset representing individuals within the United States, approximately 6,000 to 7,000 individuals are reported to have a salary exceeding $50,000.

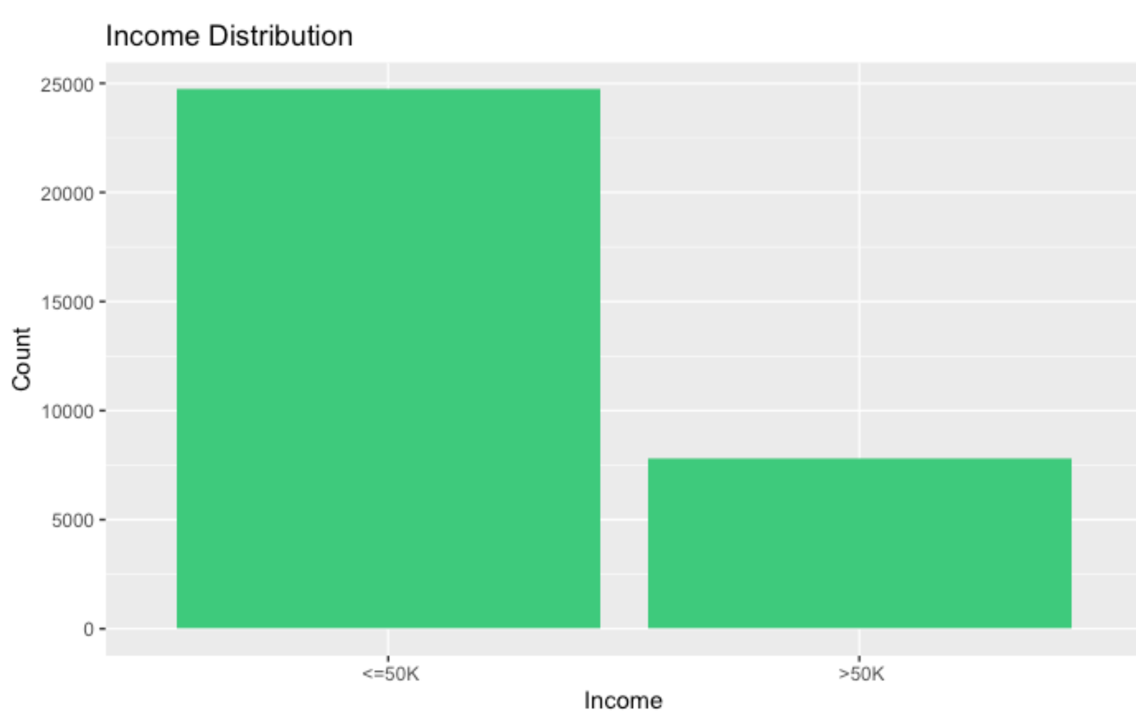
Income table

The table provides an exact count of individuals within our dataset who earn more than $50000 salary, offering precise numerical information regarding this income threshold. Additionally, th accopanying graph visually illustrates this data, presenting a clear and comprehensive representation of the distribution of individual with salaries exceeding 50k.

```
income_distribution <- as.data.frame(table(adult.data$Income))
income_distribution

##      Var1  Freq
## 1  <=50K 24719
## 2   >50K  7841

colnames(income_distribution) <- c("Income", "Count")
print(income_distribution)

##   Income Count
## 1  <=50K 24719
## 2   >50K  7841
```

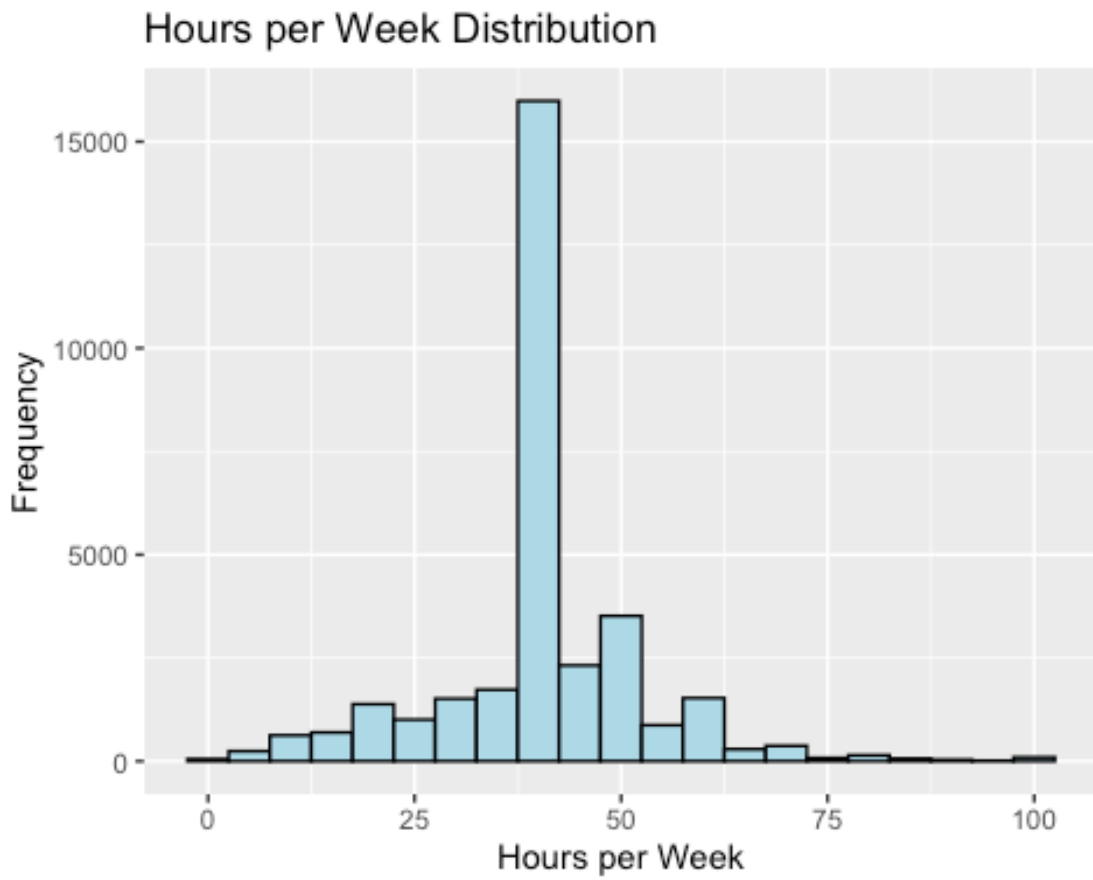
Income Distribution Graph

```
ggplot(adult.data, aes(x = Income)) +
  geom_bar(fill = "seagreen3") +
  labs(title = "Income Distribution", x = "Income", y = "Count")
```



Hours per week Distribution graph:

```
ggplot(adult.data, aes(x = Hours_per_week)) +
  geom_histogram(binwidth = 5, fill = "lightblue", color = "black") +
  labs(x = "Hours per Week", y = "Frequency", title = "Hours per Week
Distribution")
```



The distribution of hours worked per week reveals a predominant trend, with the majority of individuals dedicating their time to work within the range of 35 to 40 hours per week. This observation underscores a significant concentration of individuals whose weekly work commitment aligns with this particular time frame

Occupation By Income Graph

```
ggplot(adult.data, aes(x = Occupation, fill = Income)) +
  geom_bar() +
  labs(x = "Occupation", y = "Count", title = "Income Distribution by
Occupation") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

Individuals whose salaries surpass $50,000 are primarily concentrated within occupational roles categorized as managers and professional specialties. This suggests that a significant portion of individuals earning higher salaries tend to hold positions that require specialized skills or managerial responsibilities, reflecting a trend towards higher-paying roles in these occupational categories.

Conclusion

The Census Income dataset provides a captivating glimpse into the diverse characteristics of individuals. The age distribution visualization highlights a significant concentration of individuals aged 25 to 50, shedding light on the predominant age group within the dataset. The education distribution plot underscores the prevalence of individuals with a high school diploma, followed by those with some college education, showcasing the diverse educational backgrounds of participants.

Examining the work class distribution reveals a substantial representation of individuals employed in the private sector. Insights into marital status and gender dynamics offer a nuanced understanding of relationships within the dataset. Exploring income distribution unveils individuals earning salaries exceeding $50,000, particularly in specific occupational roles and racial groups.

In summary, these findings provide a rich understanding of the dataset's composition, paving the way for deeper analyses of socio-economic trends and patterns among individuals.